

Chapter 5

Hybrid Manifold Architecture System (HMAS)

5.1 Introduction

The central mathematical innovation of the Theory of Entropicity is the Hybrid Manifold Architecture System (HMAS).

The purpose of HMAS is to provide a unified geometric framework capable of simultaneously describing:

1. Classical information geometry.
2. Quantum information geometry.
3. Physical spacetime geometry.

Conventional differential geometry is constructed upon a single metric structure. General Relativity, for example, employs a Lorentzian metric to describe gravitation and spacetime curvature.

The Theory of Entropicity proposes that such a description is incomplete.

Information itself possesses geometry.

Quantum states possess geometry.

Spacetime possesses geometry.

If all three geometrical structures are physically meaningful, then a more comprehensive framework must incorporate them simultaneously.

HMAS is introduced precisely for this purpose.

5.2 The Foundational Geometrical Triad

The Hybrid Manifold Architecture System is built upon three distinct geometric sectors.

These sectors constitute the fundamental geometrical triad of ToE.

The sector index is defined as

$$J \in \{FR, FS, L\}.$$

The three sectors are:

FR = Fisher–Rao Sector,

FS = Fubini–Study Sector,

L = Lorentzian Sector.

Each sector possesses an independent metric structure and associated geometric quantities.

The sector index J serves as a label identifying the geometric origin of a given tensor or field.

5.3 Fisher–Rao Sector

5.3.1 Classical Information Geometry

The Fisher–Rao sector describes the geometry of classical statistical states.

A probability distribution

$$p(x|\theta)$$

is treated as a point on a statistical manifold.

The Fisher–Rao metric is

$$g_{ij}^{(FR)} = \int p(x|\theta) \frac{\partial \ln p}{\partial \theta^i} \frac{\partial \ln p}{\partial \theta^j} dx.$$

This metric quantifies statistical distinguishability.

Nearby points correspond to probability distributions that are difficult to distinguish experimentally.

Distant points correspond to distributions that are easily distinguishable.

5.3.2 Physical Interpretation

Within the Theory of Entropicity, the Fisher–Rao sector represents the geometry of classical information.

It encodes:

- uncertainty,
- statistical organization,
- information flow,
- entropy gradients.

This sector provides the classical informational substrate upon which higher-level geometric structures may emerge.

The Fisher–Rao sector is therefore interpreted not merely as a mathematical construction but as a physically significant informational geometry.

5.4 Fubini–Study Sector

5.4.1 Quantum Information Geometry

The second sector describes quantum informational structure.

For normalized quantum states

$$|\psi\rangle,$$

the Fubini–Study metric is

$$ds_{FS}^2 = 4 \left(\langle d\psi | d\psi \rangle - |\langle \psi | d\psi \rangle|^2 \right).$$

This metric defines distances on projective Hilbert space.

5.4.2 Physical Interpretation

The Fubini–Study sector captures:

- quantum distinguishability,
- quantum coherence,
- entanglement structure,
- quantum informational organization.

While the Fisher–Rao sector describes classical information, the Fubini–Study sector describes quantum information.

Together they provide a comprehensive informational foundation.

5.5 Lorentzian Sector

5.5.1 Spacetime Geometry

The Lorentzian sector corresponds to ordinary spacetime.

Its metric tensor is

$$g_{\mu\nu}^{(L)}.$$

The line element is

$$ds^2 = g_{\mu\nu}^{(L)} dx^\mu dx^\nu.$$

This is the metric of General Relativity.

All familiar gravitational phenomena emerge from this sector.

5.5.2 Physical Interpretation

The Lorentzian sector represents:

- spacetime,
- gravitation,
- causal structure,
- relativistic dynamics.

In conventional physics this sector is regarded as fundamental.

In the Theory of Entropicity it is regarded as emergent.

This distinction is crucial.

The Lorentzian sector occupies the highest level of the entropic hierarchy rather than the lowest.

5.6 Entropic Hierarchy of Emergence

The Theory of Entropicity proposes the following hierarchy:

Entropy $\longrightarrow$ Information $\longrightarrow$ Information Geometry $\longrightarrow$ Spacetime Geometry $\longrightarrow$ Gravity.

More explicitly,

$$S \longrightarrow (FR, FS) \longrightarrow L.$$

The informational sectors therefore occupy a deeper ontological level than the spacetime sector.

The Lorentzian geometry emerges from informational organization.

This represents one of the defining philosophical and mathematical commitments of the theory.

5.7 Construction of the HMAS Metric

The three sectors are unified through the Hybrid Manifold Architecture System.

The HMAS metric is defined by combining the sectoral metrics.

The fundamental decomposition is

$$g_{\mu\nu} = g_{\mu\nu}^{(FR)} + g_{\mu\nu}^{(FS)} + g_{\mu\nu}^{(L)}.$$

This relation is known as the Obidi Addition Rule for the metric.

It expresses the principle that the observable geometry contains contributions from all sectors.

5.7.1 Sectoral Interpretation

Each term contributes distinct information:

$$g_{\mu\nu}^{(FR)}$$

encodes classical informational geometry.

$$g_{\mu\nu}^{(FS)}$$

encodes quantum informational geometry.

$$g_{\mu\nu}^{(L)}$$

encodes spacetime geometry.

The total metric therefore contains both informational and gravitational content.

5.8 HMAS Determinant Structure

Because HMAS contains multiple sectors, its volume structure differs from that of conventional geometry.

The determinant factorizes according to

$$|g| = |g_{FR}| |g_{FS}| |g_L|.$$

This expression follows from the Obidi Multiplication Rule.

The corresponding invariant volume element becomes

$$dV_{HMAS} = \sqrt{|g_{FR}|} \sqrt{|g_{FS}|} \sqrt{|g_L|} d^4x.$$

This generalized measure plays a central role in the Local Obidi Action.

5.9 Sectoral Curvature

Each sector possesses independent curvature quantities.

For every sector J ,

$$R^{(J)}$$

denotes the Ricci scalar,

$$R_{\mu\nu}^{(J)}$$

denotes the Ricci tensor,

and

$$G_{\mu\nu}^{(J)}$$

denotes the Einstein tensor.

Specifically,

$$G_{\mu\nu}^{(J)} = R_{\mu\nu}^{(J)} - \frac{1}{2} g_{\mu\nu}^{(J)} R^{(J)}.$$

Each geometry therefore possesses its own gravitational-like structure.

5.10 Sectoral Independence

An important assumption of HMAS is that sectors remain geometrically distinct.

A tensor belonging to one sector is not automatically identified with a tensor belonging to another.

Thus

$$g_{\mu\nu}^{(FR)} / g_{\mu\nu}^{(FS)} \neq g_{\mu\nu}^{(L)}.$$

Each sector retains independent geometric meaning.

Communication between sectors occurs through the entropy field and through the dynamical structure of the Obidi Action.

5.11 Information Gravity

One of the most important conceptual consequences of HMAS concerns the notion of information gravity.

The reasoning proceeds as follows.

Information possesses geometry.

The Fisher–Rao metric and Fubini–Study metric establish this rigorously.

Geometry possesses gravitational significance.

General Relativity establishes this rigorously.

Therefore,

$$\text{Information Geometry} \longrightarrow \text{Information Gravity}.$$

This conclusion constitutes one of the foundational insights motivating the Theory of Entropicity.

The informational sectors are therefore not passive mathematical constructs.

They possess dynamical significance capable of influencing emergent spacetime geometry.

5.12 The Entropic Origin of Geometry

The ultimate objective of HMAS is to provide a framework through which geometry itself emerges.

The theory proposes that entropy organizes information.

Information acquires geometry.

Geometry acquires curvature.

Curvature manifests as gravity.

This chain may be written as

$$S \longrightarrow I \longrightarrow G_I \longrightarrow G_L \longrightarrow \text{Gravity}.$$

where

$$G_I$$

denotes informational geometry and

$$G_L$$

denotes Lorentzian geometry.

In this sense, spacetime becomes a macroscopic manifestation of deeper informational organization.

5.13 HMAS as the Geometric Foundation of ToE

The Hybrid Manifold Architecture System serves as the geometric foundation upon which all subsequent developments of the Theory of Entropicity are constructed.

Without HMAS there can be:

- no Obidi Convention,
- no hierarchical tensor calculus,
- no sectoral action principle,
- no entropic field equations.

HMAS therefore occupies a position analogous to that of spacetime manifolds in General Relativity.

However, whereas General Relativity employs a single geometry, HMAS employs a hierarchy of interacting geometries.

This extension constitutes one of the principal innovations of the Theory of Entropicity.

5.14 Transition to the Obidi Convention

The introduction of HMAS immediately raises a mathematical problem.

How can tensors belonging to multiple geometric sectors be represented within a single formalism?

Standard tensor notation is insufficient because it assumes only one geometric structure.

A new tensor language is therefore required.

The next chapter introduces the Obidi Convention, a hierarchical indexing framework specifically designed to describe tensors, fields, and dynamical equations within the Hybrid Manifold Architecture System.